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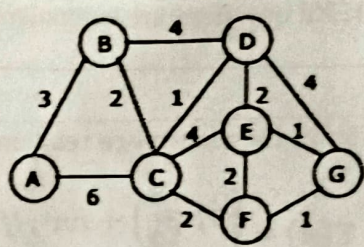
Subject Name/ Code: Design and analysis of Algorithms (IT-215/CSE-215)

MM:60

Time Allotted: 180 min

NOTE: 1. All questions are compulsory. 2. Mark answer number clearly.

1. A	Solve the recurrence relation using back substitution method. $T(n) = \begin{cases} 7T\left(\frac{n}{2}\right) + bn^2, & \text{if } n > 2, b > 0 \\ c & n \leq 2 \end{cases}$ Solve the recurrence and deduce the running time $T(n)$ is asymptotic Θ-notation. Please note that, you must prove both ends to drive the asymptotic Θ-bound.	5
B.	Consider the following recurrence to write down the code. $T(0) = T(1) = 2$ $T(n) = \sum_{k=1}^n 2 \cdot T(k) \cdot T(k-1) \text{ for } n > 1$ Is possible to improve the code for better time complexity. If yes, then explain the optimised solution with space and time complexity.	5
2. A	Given array $A[]$ with sliding window of size w which is moving from the very left of the array to the very right. Assume that we can only see the w numbers in the window. Each time the sliding window moves rightwards by one position. Find the maximum sum in sliding window. Also, discuss the time complexity.	5
B.	Consider a variant of the Merge Sort algorithm where instead of dividing the array into two approximately equal parts, it is divided into three equal parts. Each part is recursively sorted, and then the three sorted arrays are merged together. - Outline the modified MERGE-THREE (A, p, q, r, s) procedure, where A is an array and p, q, r, and s are indices into the array such that $p \leq q < r < s$. - Discuss the possible benefits or drawbacks of such a division, especially concerning the merge step. - Provide a conjecture about the time complexity of this modified algorithm compared to the standard merge sort. Is it better, worse, or the same, explain?	
3. A	Design an efficient algorithm to find the closest value to a given target number in a stored array of n distinct integers. [Example: Array: 2,5,6,7,8,8,9 Target number: 5 Output: 5 Target number: 11 Output: 9]	5

	Target number: 4 Output: 5]																															
	Write down a clean pseudocode for your algorithm. Also analyse the time complexity of your algorithm.																															
B.	Write a Huffman coding algorithm and their complexity analysis. Construct a Huffman code for the following set of symbols. Also find the average length of the Huffman code and Encode the word HIMP#XERAS. <table><tr><td>Symbols</td><td>#</td><td>A</td><td>E</td><td>H</td><td>I</td><td>L</td><td>M</td><td>P</td><td>R</td><td>S</td><td>X</td></tr><tr><td>Frequency</td><td>4</td><td>2</td><td>5</td><td>1</td><td>2</td><td>2</td><td>2</td><td>2</td><td>1</td><td>2</td><td>1</td></tr></table>	Symbols	#	A	E	H	I	L	M	P	R	S	X	Frequency	4	2	5	1	2	2	2	2	1	2	1	5						
Symbols	#	A	E	H	I	L	M	P	R	S	X																					
Frequency	4	2	5	1	2	2	2	2	1	2	1																					
4. A.	What is the role of 'min' cost edge in the graph to find minimum cost spanning tree using Krushkal algorithm? Design an algorithm for the same and find its time complexity. Use Prim's algorithm to find a minimal spanning tree for the graph shown below starting with the vertex A. 	5																														
B.	State the Job Sequencing with deadlines problem. Consider the below table for jobs given with profit and deadline. Find the maximum profit earned. <table><tr><td>Jobs</td><td>J1</td><td>J2</td><td>J3</td><td>J4</td><td>J5</td><td>J6</td><td>J7</td><td>J8</td><td>J9</td></tr><tr><td>Profit</td><td>15</td><td>20</td><td>30</td><td>18</td><td>18</td><td>10</td><td>23</td><td>16</td><td>25</td></tr><tr><td>Deadline</td><td>7</td><td>2</td><td>5</td><td>3</td><td>4</td><td>5</td><td>2</td><td>7</td><td>3</td></tr></table>	Jobs	J1	J2	J3	J4	J5	J6	J7	J8	J9	Profit	15	20	30	18	18	10	23	16	25	Deadline	7	2	5	3	4	5	2	7	3	5
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5. A.	Consider a very long, straight river which moves from north to south. Assume there are n cities on both sides of the river: n cities on the left of the river and n cities on the right side of the river. Also, assume that these cities are numbered from 1 to n but the order is not known. Now we want to connect as many left-right pairs of cities as possible with bridges such that no two bridges cross. When connecting cities, we can only connect city i on the left side to city i on the right side. Explain the time complexity of the solution.	5																														
B	Given five cities with the following distances (in km): A to B = 10, A to C = 15, A to D = 20, A to E = 25, B to C = 35, B to D = 30, B to E = 20, C to D = 10, C to E = 15, and D to E = 5, what is the minimum distance a traveling salesman must cover to visit all cities exactly once and return to the starting city? Determine the shortest possible route from node A, that visits each city once and ends at the starting point.	5																														
6. A.	Construct a system with multiple devices connected parallel in three stages. The costs of the devices are \$30, \$15 and \$20 respectively. The cost of the system is to be no more than \$105. The reliability of each device type is 0.9, 0.8 and 0.5 respectively.	5																														
B.	Four matrices M1, M2, M3 and M4 of dimensions pxq, qxr, rxs and sxt respectively can be multiplied is several ways with different number of total scalar multiplications. For example, when multiplied as ((M1 X M2) X (M3 X M4)), the total number of multiplications is pqr + rst + prt. When multiplied as (((M1 X M2) X M3) X M4), the total number of scalar multiplications is pqr + prs + pst. If p = 10, q = 100, r = 20, s = 5 and t = 80, then find the number of scalar multiplications needed.	5																														